

2. (a) When lithium reacts with water in a large beaker hydrogen gas is released.

Lithium hydroxide solution is also formed. This turns universal indicator purple.

- (i) Tick (✓) the box next to the description of what is seen when lithium reacts with water in a large beaker. [1]

lithium melts into a ball and sinks

☐

lithium fizzes and moves around the surface of the water

☐

lithium catches fire and burns with a blue flame

☐

- (ii) Tick (✓) the box that describes lithium hydroxide solution. [1]

neutral

☐

acid

☐

alkali

☐

- (iii) Lithium hydroxide contains  $\text{Li}^+$  and  $\text{OH}^-$  ions.

**Circle** the correct formula for lithium hydroxide. [1]

**LiOH**

**LiOH**

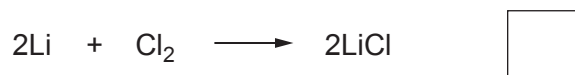
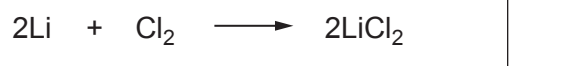
**Li(OH)<sub>2</sub>**

**Li<sub>2</sub>OH**



(b) Lithium reacts with chlorine to form lithium chloride.

(i) Tick (✓) the box next to the correct balanced equation for the reaction. [1]



(ii) Anwen was asked to use a flame test and a silver nitrate test to identify lithium chloride.

**Circle** the expected observation for each test. [2]

**Flame test**

green flame

red flame

lilac flame

**Silver nitrate test**

yellow precipitate

blue precipitate

white precipitate

(c) Lithium reacts with oxygen to form lithium oxide.

Tick (✓) the box next to the calculation used to find the relative formula mass ( $M_r$ ) of lithium oxide,  $\text{Li}_2\text{O}$ . [1]

$$A_r(\text{Li}) = 7$$

$$A_r(\text{O}) = 16$$

$$7 + 7 + 16$$

☐

$$7 + 16$$

☐

$$7 + 7 + 16 + 16$$

☐

$$7 + 16 + 16$$

☐
